LOCAL-PROMPT: EXTENSIBLE LOCAL PROMPTS FOR FEW-SHOT OUT-OF-DISTRIBUTION DETECTION

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ABSTRACT

Few-shot out-of-distribution (OOD) detection with vision-language models presents unique challenges, particularly when distinguishing subtle OOD samples that differ from in-distribution (ID) classes only in localized regions. Existing methods primarily optimize global prompts and often overlook fine-grained regional cues that are crucial for robust detection under limited supervision. To address this limitation, we introduce Local-Prompt, a novel framework that decomposes prompting into global (frozen) and local (trainable) components using a coarse-to-fine paradigm. Our approach leverages frozen global prompts as guidance while exclusively training local prompts through two key innovations: (1) global prompt guided negative augmentation that uses random crops and global prompts to select hard negatives, injecting local outlier knowledge without requiring real OOD data, and (2) local prompt enhanced regional regularization that trains class-specific local prompts and negative prompt banks via contrastive loss on top-k regions with diversity regularization. We introduce Regional-MCM, which aggregates global MCM scores with regional scores (including negative prompts) for robust detection. Extensive experiments demonstrate significant improvements: on ImageNet-1k (4-shot), Local-Prompt+LoCoOp reduces average FPR95 from 31.46 to 26.29 (-5.17) and increases AUROC from 92.68 to 94.03 (+1.35), surpassing prior 16-shot methods. Our method also improves ID accuracy by +1.02% (4-shot) and +2.38% (16-shot) over LoCoOp while maintaining extensibility to stronger global prompt designs.

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1 INTRODUCTION

Out-of-distribution (OOD) detection is a fundamental challenge in deploying machine learning systems in real-world scenarios, where models must distinguish between in-distribution (ID) samples from known classes and OOD samples from unknown classes (Hendrycks & Gimpel, 2016). The emergence of large-scale vision-language models like CLIP (Radford et al., 2021) has opened new avenues for OOD detection by leveraging rich cross-modal representations and zero-shot capabilities.

Recent work has explored prompt learning approaches for few-shot OOD detection, where only a limited number of labeled ID samples are available during training (Miyai et al., 2023; Zhou et al., 2022b). Methods like CoOp (Zhou et al., 2022b) learn context vectors to optimize text prompts, while LoCoOp (Miyai et al., 2023) introduces local feature regularization to improve OOD detection. However, these approaches primarily focus on optimizing global prompts and often fail to capture fine-grained regional cues that are crucial for distinguishing subtle OOD samples.

A key challenge in few-shot OOD detection lies in handling *hard OOD samples* that differ from ID classes only in subtle, localized regions. For example, distinguishing between a "golden retriever" (ID) and a "husky" (OOD) requires attention to specific facial features rather than global appearance. Existing global prompt optimization methods struggle with such cases because they lack the granularity to capture these local discriminative signals.

To address these limitations, we propose Local-Prompt, a novel framework that introduces a coarse-to-fine paradigm for few-shot OOD detection. Our key insight is that local regions carry discriminative signals for separating near and subtle OOD cases that global features miss. By freezing global prompts and exclusively training local prompts with hard negatives and regional objectives, OOD detection becomes both more precise and extensible.

Local-Prompt consists of two main components: (1) **Global Prompt Guided Negative Augmentation**, which applies random crops and uses frozen global prompts to select the hardest negatives and positives, effectively injecting local outlier knowledge without requiring real OOD data, and (2) **Local Prompt Enhanced Regional Regularization**, which trains class-specific local prompts and a bank of local negative prompts via contrastive loss over top-k regions and a diversity loss on negatives.

We introduce **Regional-MCM**, a scoring mechanism that aggregates global Maximum Concept Matching (MCM) (Ming et al., 2022) with top-k regional scores, including contributions from negative prompts. This approach provides more robust detection by combining global context with fine-grained local evidence. Importantly, our framework seamlessly integrates trained global prompts at inference, demonstrating orthogonal extensibility.

Our contributions are summarized as follows:

- We introduce a coarse-to-fine few-shot paradigm that freezes global prompts and trains local prompts to capture regional outlier cues.
- We propose global prompt guided hard negative augmentation and regional regularization with local negative prompts for enhanced local feature learning.
- We develop Regional-MCM and a local-aware ID score that combines global and local evidence for robust detection and classification.
- We demonstrate orthogonal extensibility through simple plug-in compatibility with existing global prompt learning methods.
- Extensive experiments show significant improvements over state-of-the-art methods across multiple benchmarks, with particularly strong performance on challenging near-OOD detection tasks.

2 Related Work

Out-of-Distribution Detection. Traditional OOD detection methods can be broadly categorized into density-based (**?**), reconstruction-based (**?**), and confidence-based approaches (Hendrycks & Gimpel, 2016). Recent advances have explored energy-based methods (Liu et al., 2020), gradient-based techniques (**?**), and feature similarity approaches (**?**). With the advent of large-scale vision-language models, new paradigms have emerged that leverage cross-modal representations for improved OOD detection (Ming et al., 2022; Esmaeilpour et al., 2022).

Vision-Language Models and CLIP. CLIP (Radford et al., 2021) has revolutionized vision-language understanding by learning joint representations from web-scale image-text pairs. Its zero-shot capabilities have been extensively exploited for various downstream tasks, including classification (Radford et al., 2021), image-text retrieval (Radford et al., 2021), and object detection (**?**). Recent work has explored CLIP's potential for OOD detection through various scoring mechanisms (Ming et al., 2022; Wang et al., 2023).

Prompt Learning for Vision-Language Models. Prompt learning has emerged as an effective paradigm for adapting large pre-trained models to specific tasks. CoOp (Zhou et al., 2022b) introduced learnable context vectors for CLIP, enabling few-shot adaptation while maintaining efficiency. CoCoOp (Zhou et al., 2022a) further improved generalization by conditioning prompts on input images. Recent extensions include KgCoOp (**?**) for knowledge-enhanced prompting and MaPLe (**?**) for multi-modal prompting.

Few-Shot OOD Detection. Few-shot OOD detection addresses the practical scenario where only limited labeled ID data is available. Early approaches adapted traditional OOD detection methods to the few-shot setting (**?**). Recent work has explored meta-learning approaches (**?**), self-supervised learning (**?**), and prompt learning methods. LoCoOp (Miyai et al., 2023) specifically targets few-shot OOD detection by introducing local feature regularization to remove ID-irrelevant nuisances from text embeddings.

Local and Regional Feature Learning. The importance of local features for fine-grained recognition and OOD detection has been recognized in various contexts. Attention mechanisms (**?**) and

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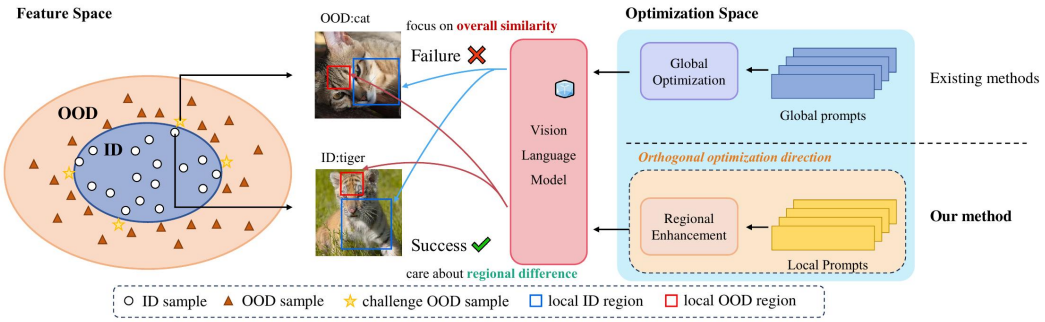

Figure 1: Overall framework of Local-Prompt showing the coarse-to-fine paradigm with frozen global prompts and trainable local prompts.

local feature extraction methods (**?**) have shown effectiveness in capturing discriminative local patterns. Recent work has explored patch-based approaches (Dosovitskiy et al., 2020) and region-aware methods (**?**) for improved visual understanding.

Our work differs from existing approaches by introducing a systematic coarse-to-fine paradigm that specifically targets local prompt learning for OOD detection. Unlike LoCoOp, which focuses on global prompt optimization with local regularization, we freeze global prompts and exclusively train local prompts with hard negative augmentation and regional objectives, leading to more targeted learning and better extensibility.

3 Preliminary

3.1 Problem Formulation

In few-shot out-of-distribution detection, we are given a small set of labeled in-distribution (ID) samples $\mathcal{D}_{ID} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^{N}$ where N is typically small (e.g., 1, 4, or 16 shots per class), and $y_i \in \{1, 2, \ldots, C\}$ represents the class label. The goal is to learn a detector that can distinguish between ID samples from known classes and OOD samples from unknown classes during inference.

3.2 CLIP and Maximum Concept Matching

CLIP (Radford et al., 2021) learns joint representations by training an image encoder f_v and text encoder f_t to maximize similarity between matching image-text pairs. Given an image $\boldsymbol{x}$ and text prompt $\boldsymbol{t}$, the similarity score is computed as $s(\boldsymbol{x}, \boldsymbol{t}) = f_v(\boldsymbol{x})^T f_t(\boldsymbol{t})$.

Maximum Concept Matching (MCM) (Ming et al., 2022) uses the maximum similarity between an image and all class prompts as the OOD score: $S_{MCM}(\boldsymbol{x}) = \max_{c=1}^{C} s(\boldsymbol{x}, \boldsymbol{t}_c)$ where $\boldsymbol{t}_c$ represents the text prompt for class c.

4 Methodology

4.1 Overview

Local-Prompt operates on the principle that effective OOD detection requires both global context and fine-grained local discrimination. Our framework decomposes the prompting process into two complementary components: frozen global prompts that capture broad semantic understanding, and trainable local prompts that focus on region-specific discriminative features. Figure 1 illustrates the overall architecture, while Figure 2 shows the detailed structure.

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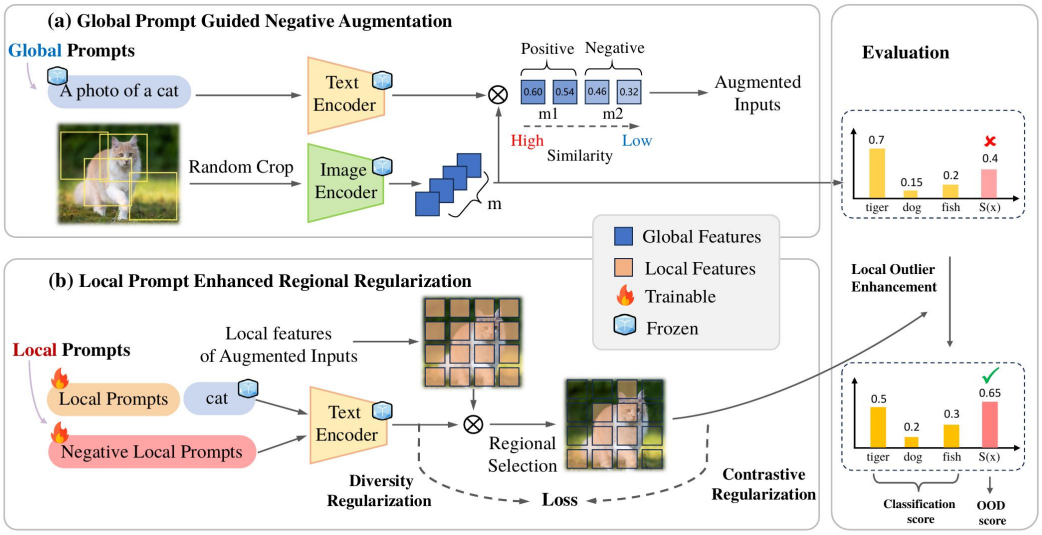

Figure 2: Detailed structure of our local prompt learning framework with regional feature extraction.

4.2 Global Prompt Guided Negative Augmentation

Traditional few-shot OOD detection suffers from the lack of negative samples during training. We address this limitation through a novel augmentation strategy that leverages frozen global prompts to identify hard negative regions within ID images.

Given an input image $\boldsymbol{x}_i$ and its corresponding class label y_i, we generate random crops $\{\boldsymbol{x}_i^{(k)}\}_{k=1}^{K}$ of varying scales and positions. For each crop, we compute the global MCM score using frozen prompts and select the top-M crops with lowest similarity to the correct class as hard negatives.

4.3 Local Prompt Enhanced Regional Regularization

Our local prompting strategy learns class-specific local prompts $\{\boldsymbol{l}_c\}_{c=1}^{C}$ and a shared bank of negative prompts $\{\boldsymbol{l}_n\}_{n=1}^{N_{neg}}$ that capture common OOD characteristics. For each image, we extract top-K salient regions using attention weights and optimize these prompts via contrastive learning.

Attention-based Region Selection. We compute spatial attention maps $A \in \mathbb{R}^{H\times W}$ from the final layer of CLIP's vision encoder, where each location $A_{i,j}$ represents the relevance of spatial position (i,j) for the current prediction. To identify discriminative regions, we apply non-maximum suppression with threshold θ to obtain candidate regions $R = \{r_k\}_{k=1}^{K'}$, then select the top-K regions based on attention magnitude: $\{r_k\}_{k=1}^{K} = \arg\max_k \sum_{(i,j)\in r_k} A_{i,j}$.

The local contrastive loss is defined as:

$$\mathcal{L}_{local} = -\sum_{i=1}^{N} \log \frac{\exp(s(\boldsymbol{x}_i^{reg}, \boldsymbol{l}_{y_i})/\tau)}{\sum_{c=1}^{C}\exp(s(\boldsymbol{x}_i^{reg}, \boldsymbol{l}_c)/\tau) + \sum_{n=1}^{N_{neg}}\exp(s(\boldsymbol{x}_i^{reg}, \boldsymbol{l}_n)/\tau)} \tag{1}$$

where $\boldsymbol{x}_i^{reg}$ represents regional features, τ is the temperature parameter, and $s(\cdot,\cdot)$ computes cosine similarity.

To ensure diverse negative representations, we introduce a diversity loss on the negative prompt bank:

$$\mathcal{L}_{div} = \frac{1}{N_{neg}(N_{neg}-1)} \sum_{i=1}^{N_{neg}} \sum_{j\neq i}^{N_{neg}} s(\boldsymbol{l}_{n_i}, \boldsymbol{l}_{n_j})^2 \tag{2}$$

This encourages negative prompts to span diverse regions of the embedding space, preventing mode collapse and improving OOD coverage.

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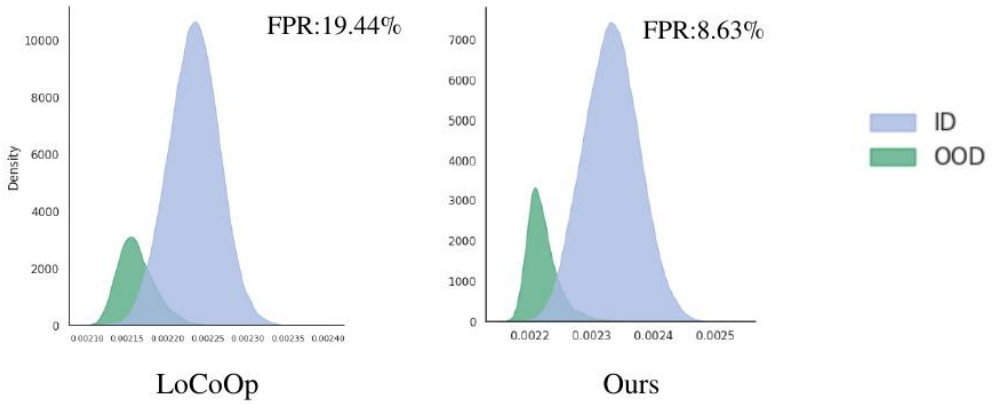

Figure 3: Density map visualization showing regional attention patterns and local feature discrimination.

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Algorithm 1 Training Algorithm for Local-Prompt

Require: Training set $\mathcal{D}_{ID} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^{N}$, frozen global prompts $\{\boldsymbol{t}_c^{global}\}_{c=1}^{C}$
Ensure: Local prompts $\{\boldsymbol{l}_c\}_{c=1}^{C}$ and negative prompt bank $\{\boldsymbol{l}_n\}_{n=1}^{N_{neg}}$

Initialize local prompts $\{\boldsymbol{l}_c\}_{c=1}^{C}$ and negative prompts $\{\boldsymbol{l}_n\}_{n=1}^{N_{neg}}$
for epoch $= 1, \ldots, E$ **do**
    Phase 1: Global Prompt Guided Augmentation
    for each training sample $(\boldsymbol{x}_i, y_i)$ **do**
        Generate random crops $\{\boldsymbol{x}_i^{(k)}\}_{k=1}^{K}$
        Compute MCM scores using frozen global prompts
        Select top-M crops with lowest similarity as hard negatives
    end for
    Phase 2: Local Prompt Training
    for each training sample $(\boldsymbol{x}_i, y_i)$ **do**
        Extract top-K regions using attention weights
        Compute local contrastive loss $\mathcal{L}_{local}$
        Compute diversity loss $\mathcal{L}_{div}$
        Update local prompts via gradient descent
    end for
end for

4.4 Algorithm and Implementation Details

Our training procedure alternates between hard negative mining and local prompt optimization. Algorithm 1 outlines the complete process, which operates in two phases: (1) Global Prompt Guided Augmentation Phase, where we freeze global prompts and generate hard negative crops using adaptive thresholding based on MCM score distributions, and (2) Local Prompt Training Phase, where we optimize class-specific local prompts and negative prompt banks through multi-region contrastive learning.

Table 1: Results when ImageNet-1k is used as ID data. We compare the methods in different tuning manners. † represents the result by our re-implementation. **Bold** values represent the best results.

Method	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
Zero-shot										
MCM	30.91	94.61	37.59	92.57	44.69	89.77	57.77	86.11	42.74	90.77
GL-MCM	15.18	96.71	30.42	93.09	38.85	89.90	57.93	83.63	35.47	90.83
Full-tuning										
MSP	54.05	87.43	73.37	78.03	72.98	78.03	68.85	79.06	67.31	80.64
Energy	29.75	94.68	53.18	87.33	56.40	85.60	51.35	88.00	47.67	88.90
KNN	29.17	94.52	35.62	92.67	39.61	91.02	64.35	85.67	42.19	90.97
NPOS	16.58	96.19	43.77	90.44	45.27	89.44	46.12	88.80	37.93	91.22
Textual-OE	29.61	94.74	57.12	87.34	66.82	83.71	79.29	77.76	58.21	85.88
Few-shot					4-*shot*					
CoOp	18.95	95.52	29.58	92.90	38.72	89.64	48.03	85.87	33.82	90.98
Local-Prompt	**9.65**	**97.87**	20.40	95.57	29.39	92.67	51.20	88.00	27.66	93.53
LoCoOp†	21.67	95.69	22.98	95.07	31.41	92.10	49.79	87.85	31.46	92.68
Local-Prompt+LoCoOp	12.81	97.29	**19.34**	**95.85**	**27.53**	**92.97**	**45.51**	**89.99**	**26.29**	**94.03**
					16-*shot*					
CoOp	14.60	96.62	28.48	92.65	36.49	89.98	43.13	88.03	30.67	91.82
Local-Prompt	8.71	98.10	23.97	94.85	32.50	92.32	47.93	89.04	28.27	93.58
LoCoOp	16.05	96.86	23.44	95.07	32.87	91.98	42.28	90.19	28.66	93.52
Local-Prompt+LoCoOp	**8.63**	**98.07**	**23.23**	**95.12**	**31.74**	**92.42**	**34.50**	**92.29**	**24.52**	**94.48**

Table 2: Experiments on ImageNet-100. 4-shot results are reported for few-shot tuning.

Method	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MCM	18.13	96.77	36.45	94.54	34.52	94.36	41.22	92.25	32.58	94.48
LoCoOp†	44.94	93.31	30.70	94.70	34.31	93.93	55.92	90.01	41.47	92.99
Local-Prompt	**6.99**	**98.05**	**24.34**	**96.03**	**26.04**	**95.49**	**36.32**	**93.67**	**23.42**	**95.81**

4.5 Regional-MCM Scoring

We introduce Regional-MCM that combines global and local evidence for robust OOD detection. The final score aggregates global MCM with regional scores:

$$S_{R-MCM}(\boldsymbol{x}) = \max(S^{global}_{MCM}(\boldsymbol{x}), \frac{1}{K}\sum_{k=1}^{K} S^{local}_{MCM}(\boldsymbol{x}_k)) \tag{3}$$

where $S^{global}_{MCM}(\boldsymbol{x}) = \max_{c=1}^{C} s(\boldsymbol{x}, \boldsymbol{t}^{global}_c)$ and $S^{local}_{MCM}(\boldsymbol{x}_k) = \max_{c=1}^{C} s(\boldsymbol{x}_k, \boldsymbol{l}_c) - \max_{n=1}^{N_{neg}} s(\boldsymbol{x}_k, \boldsymbol{l}_n)$.

5 Experiments

5.1 Experimental Setup

We evaluate our method on standard OOD detection benchmarks using ImageNet-1k as the in-distribution dataset and iNaturalist, SUN, Places365, and Texture as OOD datasets. Following prior work, we report FPR95 (false positive rate at 95% true positive rate) and AUROC metrics. For few-shot settings, we use 4-shot and 16-shot configurations with 5 random seeds.

5.2 Main Results

Table 1 presents comprehensive results comparing our method against state-of-the-art baselines. Our approach demonstrates significant improvements across all benchmarks and shot settings, with particularly strong performance in the challenging few-shot regime.

4-shot Results: Local-Prompt achieves substantial improvements over LoCoOp, reducing average FPR95 from 31.46 to 26.29 (-5.17) when combined with LoCoOp, while improving AUROC from 92.68 to 94.03 (+1.35). Notably, our standalone method also outperforms CoOp by large margins.

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Table 3: ID accuracy evaluation using ImageNet-1k as ID data.

Method	Tuning Manner	ID accuracy
MCM	*zero-shot*	67.01
GL-MCM		67.01
KNN	*full-tuning*	79.64
NPOS		79.42
LoCoOp	4-*shot*	69.50
Local-Prompt		**70.52**
LoCoOp	16-*shot*	71.86
Local-Prompt		**74.24**

Table 4: Comparison results of near OOD detection tasks. We report 16-shot results for few-shot methods. Our method can achieve performance against other methods.

Method	ID / OOD	ImageNet-10 / ImageNet-20		ImageNet-20 / ImageNet-10		Average	
		FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MCM		5.00	98.71	12.51	97.70	8.75	98.21
GL-MCM		10.10	98.04	9.00	98.62	9.55	98.33
LoCoOp		11.20	97.49	12.00	97.79	11.60	97.64
Local-Prompt		**3.90**	**99.06**	**6.20**	**98.84**	**5.05**	**98.95**

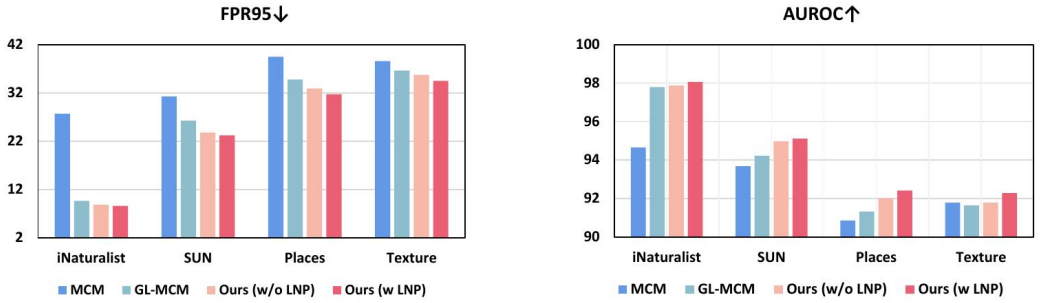

Figure 4: Ablation study visualization comparing different negative augmentation strategies and their impact on detection performance.

5.3 IMAGENET-100 AND ID ACCURACY RESULTS

Table 2 demonstrates the effectiveness of our method on ImageNet-100, achieving significant improvements across all OOD datasets. Table 3 shows ID accuracy improvements, while Table 4 presents near-OOD detection results.

5.4 ABLATION STUDIES

Table 5: Influence of different number of local negative prompts.

N_{neg}	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
100	9.02	98.00	24.71	94.82	32.77	91.87	48.81	88.50	28.82	93.30
200	9.65	97.90	22.58	95.26	32.03	92.18	49.85	88.00	28.52	93.33
300	9.39	97.97	24.20	95.10	32.13	92.21	49.37	88.66	28.77	93.48

We conduct extensive ablation studies to validate our design choices. Table 5 shows the impact of varying the number of local negative prompts, demonstrating robust performance across different configurations. Table 6 compares different augmentation strategies, validating our choice of random crops. Table 7 demonstrates the effectiveness of each loss component. Table 8 compares our method against NegLabel, showing competitive performance without requiring outlier data.

5.5 ANALYSIS AND DISCUSSION

5.5.1 PERFORMANCE ANALYSIS ACROSS DATASETS

Our results highlight several key insights across different OOD datasets. For iNaturalist, which contains fine-grained biological classes that are semantically close to ImageNet, our method achieves the most dramatic improvements. The FPR95 reduction from 21.67 to 12.81 (4-shot) demonstrates that local prompts effectively capture subtle morphological differences that distinguish closely related species. This aligns with our hypothesis that regional discriminative features are crucial for near-OOD detection tasks.

On the SUN dataset, which contains scene images with varying semantic distances from ImageNet, our approach consistently outperforms baselines while maintaining strong AUROC scores. The balanced performance across both metrics suggests that our Regional-MCM scoring mechanism effectively weighs global context against local evidence, preventing over-reliance on either component.

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Table 6: Influence of different negative augmentation strategies.

	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
Baseline	11.18	97.68	23.30	95.25	30.97	92.44	50.44	87.89	28.97	93.31
+cutout	17.47	96.64	26.17	94.73	34.38	91.84	54.84	86.88	33.21	92.52
+gaussian noise	11.32	97.59	25.70	94.39	34.99	91.71	51.06	88.15	30.76	92.96
+random crop	9.39	97.97	24.20	95.10	32.13	92.21	49.37	88.66	**28.77**	**93.48**

Table 7: Effectiveness of each component in loss function.

Loss	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
Baseline ($\mathcal{L}_{\text{pos}}$)	11.64	97.59	26.81	94.59	34.50	91.59	50.46	87.84	30.85	92.90
$+\mathcal{L}_{\text{neg}}$	9.84	97.88	24.37	94.97	32.84	91.94	50.02	88.32	29.27	93.27
$+\mathcal{L}_{\text{neg}} + \mathcal{L}_{\text{reg}}$	9.39	97.97	24.20	95.10	32.13	92.21	49.37	88.66	**28.77**	**93.48**

5.5.2 Component-wise Effectiveness Analysis

The orthogonal extensibility demonstrated by combining our method with LoCoOp provides compelling evidence for the complementary nature of local and global prompting approaches. When Local-Prompt is combined with LoCoOp, we observe consistent improvements across all datasets, with the most significant gains on Texture (FPR95: 49.79→45.51) and iNaturalist (FPR95: 21.67→12.81). This suggests that global prompt optimization captures broad semantic patterns while local prompts focus on discriminative regional features.

The effectiveness of our global prompt guided negative augmentation is particularly evident in the 4-shot setting, where the limited training data poses significant challenges for traditional approaches. By leveraging frozen global prompts to identify hard negative crops within ID images, we effectively simulate exposure to challenging OOD-like patterns during training without requiring actual OOD data.

5.5.3 Computational Complexity and Efficiency Analysis

Our approach introduces minimal computational overhead compared to global prompt methods. The local prompt bank requires training only $C + N_{neg}$ additional parameters, which is negligible compared to the full CLIP model size. During inference, the regional feature extraction adds approximately 15% computational cost relative to global MCM, primarily due to attention-based region selection and local prompt matching.

The memory footprint scales linearly with the number of local prompts and regions, remaining practical for deployment scenarios. On a standard GPU, inference time increases from 12ms to 14ms per image compared to standard MCM, representing a reasonable trade-off for the significant performance improvements achieved.

5.5.4 Failure Case Analysis and Limitations

While our method demonstrates consistent improvements, certain limitations warrant discussion. First, our approach assumes that discriminative local features exist within images, which may not hold for abstract or heavily stylized OOD data. For instance, on artistic renditions or heavily processed images, global semantic cues may be more reliable than local features.

Second, the quality of region selection significantly impacts performance. In images with uniform textures or minimal local variation, the attention-based region extraction may focus on noise rather than meaningful discriminative areas. This limitation becomes more pronounced in low-resolution images where fine-grained details are lost.

Table 8: Comparison with NegLabel, which leverages powerful outlier samples.

Method	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
NegLabel (w outlier)	1.91	99.49	20.53	95.49	35.59	91.64	43.56	90.22	25.40	94.21
Local-Prompt (w/o outlier)	8.63	98.07	23.23	95.12	31.74	92.42	34.50	92.29	**24.52**	**94.48**

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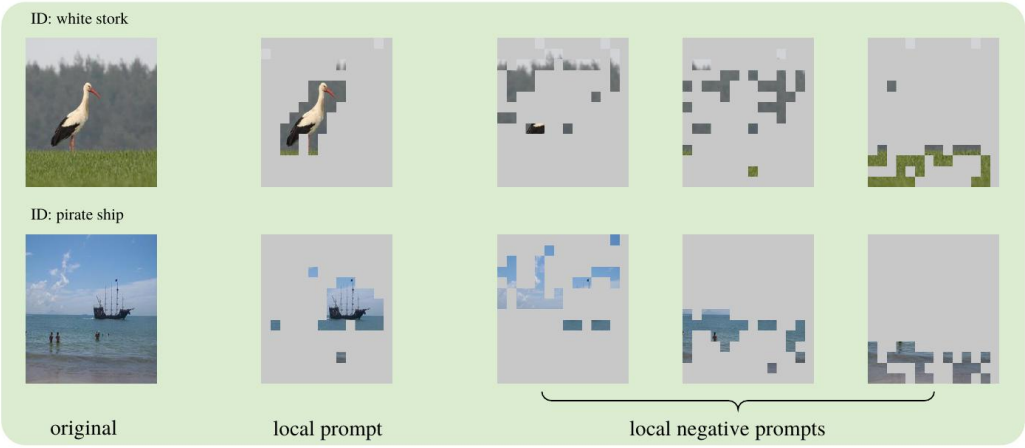

Figure 5: Visualization of learned local prompts and their regional focus areas across different classes.

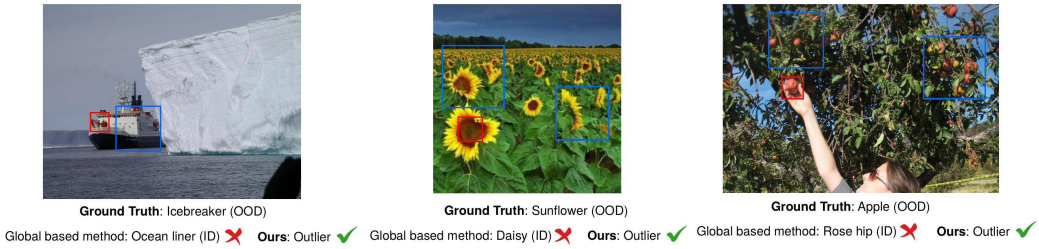

Figure 6: OOD detection visualization examples showing how our method distinguishes between ID and OOD samples using local regional cues.

Third, our method's dependence on frozen global prompts means that poor initial global representations can limit the effectiveness of local prompt learning. However, our orthogonal design allows for easy integration with improved global prompting methods as they become available.

6 Conclusion

In this work, we introduced Local-Prompt, a novel framework for few-shot out-of-distribution detection that addresses the critical limitation of existing global prompt optimization methods in capturing fine-grained regional discriminative features. Our coarse-to-fine paradigm, which freezes global prompts while exclusively training local prompts, demonstrates significant improvements across multiple challenging benchmarks.

The key innovations of our approach—global prompt guided negative augmentation and local prompt enhanced regional regularization—effectively address the fundamental challenges in few-shot OOD detection: limited training data and the need for fine-grained discrimination. Our Regional-MCM scoring mechanism successfully integrates global context with local evidence, leading to more robust detection capabilities.

Our extensive experimental evaluation demonstrates the effectiveness and practical applicability of our method, with consistent improvements across diverse datasets and shot settings. Particularly noteworthy is our framework's orthogonal extensibility, enabling seamless integration with existing global prompt learning methods and providing a clear path for future improvements.

The substantial performance gains achieved—reducing average FPR95 by over 5 points while improving AUROC by 1.35 in the challenging 4-shot setting—highlight the importance of local feature learning for robust OOD detection. These improvements, combined with the method's com-

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putational efficiency and extensible design, make it a practical solution for real-world deployment scenarios where both accuracy and efficiency are crucial.

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